

ARTICLE

Methods, Tools, and Technologies

Evaluating the robustness of generalized additive models as a tool for threshold detection in variable environments

A. Raine Detmer¹  | Eric J. Ward²  | Mary E. Hunsicker³ |
 Kelly S. Andrews²  | Michele Conrad⁴ | Bridget E. Ferriss⁵  |
 Elliott L. Hazen⁶  | Kirstin K. Holsman⁷ | Julia Indivero⁸ | Scott I. Large⁹ |
 Michael Malick¹⁰  | Kristin N. Marshall¹¹ | Stuart H. Munsch¹² |
 Kiva L. Oken¹¹  | William H. Satterthwaite¹³ | S. Kalei Shotwell⁵ |
 Andrew R. Thompson¹⁴ | Jameal F. Samhouri²

¹Department of Ecology, Evolution, and Marine Biology, University of California, Santa Barbara, California, USA

²Conservation Biology Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, Washington, USA

³Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Newport, Oregon, USA

⁴Oceanbeat Consulting, LLC, Olympia, Washington, USA

⁵Resource Ecology and Fisheries Management Division, Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, Washington, USA

⁶Ecosystem Science Division, Southwest Fisheries Science Center, National Oceanic and Atmospheric Administration, Monterey, California, USA

⁷Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, Washington, USA

⁸School of Aquatic and Fishery Sciences, University of Washington, Seattle, Washington, USA

⁹Northeast Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Woods Hole, Massachusetts, USA

¹⁰Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Port Orchard, Washington, USA

¹¹Fishery Resource Analysis and Monitoring Division, Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, Washington, USA

¹²Northwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Pasco, Washington, USA

¹³Fisheries Ecology Division, Southwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Santa Cruz, California, USA

¹⁴Ecosystem Science Division, Southwest Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, La Jolla, California, USA

Correspondence

A. Raine Detmer

Email: adetmer@ucsb.edu

Funding information

NOAA Integrated Ecosystem Assessment Program; NSF INTERN, Grant/Award Number: 2139319

Abstract

As global climate change and anthropogenic activities amplify widespread environmental variability, there is a strong need for management strategies that incorporate relationships between ecosystem components. This need is especially apparent when changes in environmental drivers cause threshold responses (abrupt, nonlinear changes) in ecosystems. Such ecological

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Ecosphere* published by Wiley Periodicals LLC on behalf of The Ecological Society of America. This article has been contributed to by U.S. Government employees and their work is in the public domain in the USA.

Handling Editor: David M. Bell

thresholds can provide useful reference points for management decisions. However, methods for detecting thresholds in empirical datasets may fail to find an existing threshold, find one that does not exist, or be biased in their estimates of threshold locations. These types of threshold misspecifications can result in high conservation and socioeconomic costs. Simulation studies can mitigate these risks by providing information about method performance across different scenarios. Here, we constructed a series of simulations to evaluate the robustness of threshold detection with generalized additive models (GAMs) when exposed to a variety of common, real-world data characteristics. GAMs generally performed best when time series were long, observation error was low, thresholds were crossed fairly frequently, and covariates were accounted for. Over realistic ranges of values, observation error and frequency of threshold crossing had stronger effects on threshold detectability than time series length. Importantly, detectability was found to depend on both the shape of the threshold relationship and the statistical definition of the threshold location. As a case study, we applied this threshold detection method to an empirical dataset relating ocean temperature and the spatial distribution of Pacific hake (*Merluccius productus*), the largest volume fishery on the US West Coast. While the data suggest no statistical evidence for a threshold relationship, our simulations indicated approximately equal chances of true and false threshold detection given currently available data. Our results provide general guidelines for where threshold detection with GAMs is likely to be robust and are useful in the context of indicator development for ecosystem-based management in a variable world.

KEYWORDS

ecosystem thresholds, ecosystem-based management, generalized additive models, Pacific hake, reference points, simulation studies

INTRODUCTION

A core principle of environmental management is delineation of a boundary distinguishing sustainable from unsustainable practices. Such boundaries, or reference points, can help decision makers modify human activities to encourage continued human use and societally desired environmental states (Samhouri et al., 2010, 2017; Selkoe et al., 2015). For example, toxicity of pollutants, exploitation rates in harvested populations, extent of drought in relation to freshwater usage, and loading of nutrients into aquatic systems all can have reference points to inform management decisions aimed at preventing adverse environmental, social, and economic impacts (Groffman et al., 2006). A key challenge is that global change can increase the probability of breaching such reference points (Gove et al., 2023; Phillips et al., 2024; Wang, 2024). Heightened concern emerges when abrupt, nonlinear change in an ecosystem occurs in relation to a more gradual

change in an environmental or human driver. Crossing of such “ecosystem thresholds” is well-documented across terrestrial, freshwater, and marine systems (Kelly et al., 2015; USGCRP, 2023) and provides objective information for complex environmental management decisions (Morrison et al., 2024). However, new meta-analyses and methodological advances call into question both the ubiquity of and the ability to identify ecosystem thresholds (Hillebrand et al., 2020; Satterthwaite et al., 2020). Moreover, they suggest a need to scrutinize quantitative methods used to detect ecosystem thresholds and more thoughtful consideration of how to apply these methods in real-world situations (Spake et al., 2022).

Many statistical methods for threshold or change point detection exist and have been advanced across a variety of disciplines. These methods include detection statistics (Carpenter et al., 2014); regression-based approaches, such as piecewise regression; and generalized additive models (GAMs; Benedetti et al., 2009),

which implement basis functions to capture nonlinear relationships. Machine learning methods including random forests, neural networks, support vector machines, and empirical dynamic modeling have also been applied to detect nonlinear thresholds and regime shifts (Aminikhanghahi & Cook, 2017; Lonschien et al., 2023; Munch et al., 2023). Relative to other fields, datasets in ecology and fishery subfields are often short, gappy, and corrupted by observation errors that are generally not known a priori. As such, simpler methods like GAMs have been popularized in these fields. GAMs are widely accessible in software including R (R Core Dev Team, 2024; Wood, 2011), are able to estimate nonlinear thresholds with the addition of only several parameters, and provide output that is relatively easy to interpret (Ficetola & Denoël, 2009; Large et al., 2013; Sonderegger et al., 2009).

Numerous factors influence the robustness of GAMs and other statistical threshold detection methods. One such factor is sample size: A priori, we would expect threshold detection to be more accurate when the total number of data points (e.g., years or spatial replicates in a survey) is higher. Another component of sample size is the frequency at which thresholds are crossed, which is determined by the frequency of observations across possible values of the driver (i.e., the sample-environmental distribution of the driver; Daily et al., 2012). If a driver only exceeds a threshold during extreme events, most (or all) data points may be clustered on one side of a threshold, reducing detectability. This challenge has hindered attempts to identify metabolic index thresholds in West Coast groundfish species, where oxygen levels rarely exceed expected threshold levels (Indivero et al., 2024). High levels of noise in the data (due to observation error and/or unknown covariates) may also negatively influence threshold detectability; for example, high interannual variability in the North Pacific has complicated efforts to detect regime shifts in this system (Wooster & Zhang, 2004).

A third factor to consider when evaluating threshold detection methods is the shape of the driver–response relationship. While the term “threshold” has been defined in various ways in the literature (Groffman et al., 2006; Spake et al., 2022), thresholds that are of utility in a management context could include any value of a driver at which the rate and/or direction of change in a response variable with respect to that driver shifts abruptly. For example, Samhouri et al. (2010) identified sigmoidal and hockey stick relationships between anthropogenic pressures and ecosystem attributes (net primary productivity: biomass ratios and Shannon diversity, respectively) in the northeastern Pacific; in both cases, these attributes remained relatively constant across a range of pressures but changed quickly once pressures exceeded threshold values. Other threshold-like relationships

include unimodal shapes, such as skewed thermal performance curves in which the performance of an organism initially increases with temperature before dropping off rapidly (Deutsch et al., 2008; Spake et al., 2022). In all cases, knowledge of where the onset of rapid change occurs provides managers with a valuable reference point for decision-making. However, beyond investigating threshold detection as a function of the rate of linear change (Daily et al., 2012), the effect of the shape of driver–response relationships on our ability to detect such change points has not been widely explored.

While empirical studies have used GAMs for threshold detection in fisheries (Hunsicker et al., 2016; Large et al., 2013; Muhling et al., 2020; Samhouri et al., 2017; Satterthwaite et al., 2020; Weber & McClatchie, 2012), a rigorous evaluation of the robustness of this method has been lacking and is needed to determine the appropriate level of confidence when using outcomes for resource management and conservation. Simulation studies can help quantify effects of the factors discussed above, providing expectations for the robustness of threshold detection methods applied to specific datasets. Simulations also allow for confronting insights gained from theoretical evaluation of statistical methods with the practicalities associated with real-world data. By tuning simulation parameters to represent an empirical dataset, simulations can provide insight into the likelihood that statistical methods will correctly detect a threshold in those data, providing managers with an additional level of confidence in decisions about incorporating threshold responses into their management strategies.

Pacific hake (*Merluccius productus*) distributions in the California Current Ecosystem provide a useful case study for exploring the statistical power of GAMs to detect thresholds from environmental datasets. Pacific hake supports the largest commercial fishery (by volume) on the US West Coast (Berger et al., 2022) and is a transboundary stock jointly monitored and managed by the United States and Canada. Management relies on a fixed allocation of total catch to the United States and Canada as specified in the Pacific Hake/Whiting Treaty, independent of how the hake population is distributed in any given year. Mismatches between the fixed allocation and variable distribution of the stock have been shown to increase future risks of fishery closures (Jacobsen et al., 2022), and managers have expressed interest in whether there is a threshold response of the spatial distribution of Pacific hake to ocean temperatures. Previous work indicates temperature influences the spatial distribution of Pacific hake in the California Current Ecosystem, whereby a greater number of adult hake may migrate farther north into Canadian waters in years with warmer water temperatures (Agostini et al., 2008; Malick et al., 2020). The existence of a threshold in the

relationship between temperature and hake distribution could offer fishery managers and the fishing industry the ability to anticipate sudden changes in hake availability to US and Canadian fisheries, providing a useful reference point when deciding whether stock allocations between the two countries should be changed based on the spatial distribution of the stock in relation to temperature forecasts. Such knowledge would be especially valuable for informing harvest strategies in the face of increasing variability in climate and ocean conditions.

In this study, we used simulations to explore how GAM threshold detection methods perform under various hypothetical scenarios (i.e., time series lengths, level of observation error, frequency at which the true threshold is crossed, and presence of covariates). We considered several generic functional forms in our simulations, allowing us to investigate whether detectability depends on the nature of the driver–response relationship. Our goal was to develop a foundational understanding of GAM performance, and we therefore focused on relatively simple scenarios (i.e., bivariate driver–response relationships, no spatial or temporal autocorrelation or nonstationary) that could be easily analyzed and interpreted, but we suggest extensions of our work in the discussion. Finally, to evaluate our ability to detect thresholds in a real-world scenario of interest to managers, we applied our simulation methods to an empirical case study on Pacific hake. This work provides a preliminary evaluation of the robustness of GAM threshold detection methods, helping to advance the utility of ecosystem thresholds as a foundation for developing ecosystem-based fishery management strategies.

METHODS

Simulation overview

To simulate driver–response data, we used one of three threshold driver–response relationships (skewed curve, sigmoidal, and hockey stick; Figure 1) and one linear relationship, with the latter serving as a null model to test false positive rates (FPRs). We focused on evaluating the identification of probable (i.e., grounded in known or hypothesized mechanisms) thresholds in simple bivariate driver–response relationships. Such relationships are simple, easy to communicate, and have been the focus of previous studies using the GAM methods we evaluate here (Large et al., 2013; Samhouri et al., 2017). However, many threshold relationships in real ecosystems likely have higher dimensionality, and we return to this point in the *Discussion*.

It is often the case that one side of a threshold represents a less desirable state, making threshold estimates biased toward this side riskier for managers. To emphasize

this asymmetry in bias, we labeled each side of our thresholds as “risk-prone” (undesirable response values) or “risk-averse” (desirable response values). We assumed the side of the threshold with the majority of observations was risk-averse, representing a case where the response is normally in a desirable state but extreme driver values can cause an abrupt shift to undesirable conditions (but we acknowledge other scenarios are possible, e.g., if the system being monitored was currently in the undesirable state). For cases in which there is no clear “desirable” or “undesirable” side of a threshold, our labels simply provide a reference for the direction of bias in threshold estimates.

Response values were generated by applying one of our four relationships (skew, sigmoidal, hockey stick, or linear) to simulated driver data and corrupting the predicted responses with random observation error. For some scenarios, we simulated response values as a function of both the driver and an additional covariate, where the additional covariate had either a linear or nonlinear (exponential) relationship with the response. This represents a situation in which the response not only has a threshold relationship with the driver but is also influenced by another variable (e.g., species distributions might have threshold relationships with oxygen, with temperature as an additional covariate; Essington et al., 2022). For cases in which data were simulated with the driver and additional covariate, we considered two scenarios: (1) the additional covariate is unknown (latent) and not included in the estimation models and (2) the additional covariate is known and observed, with simulated values included in the estimation models. For simplicity, we assumed the driver and additional covariate were independent (no correlation), and our results therefore likely represent a best-case scenario in terms of the ability of GAMs to estimate thresholds when the response is influenced by additional covariates. After simulating the data, we specified linear regression and GAMs and fit these estimation models to each of the simulated driver–response time series. For scenarios in which the additional covariate was assumed to be observed, we fit a suite of linear and GAM models that represented different treatments of the driver and additional covariate. For each simulated dataset, we used the Akaike information criterion (AIC) score (or corrected AIC [AIC_c], for sample sizes <40) to select the best-supported estimation model. A full description of the simulation procedure is provided in Appendix S1.

Threshold statistics

When the best-supported model was nonlinear (with a difference in AIC_c between the linear and nonlinear models [ΔAIC_c] of at least 2), we searched for a threshold

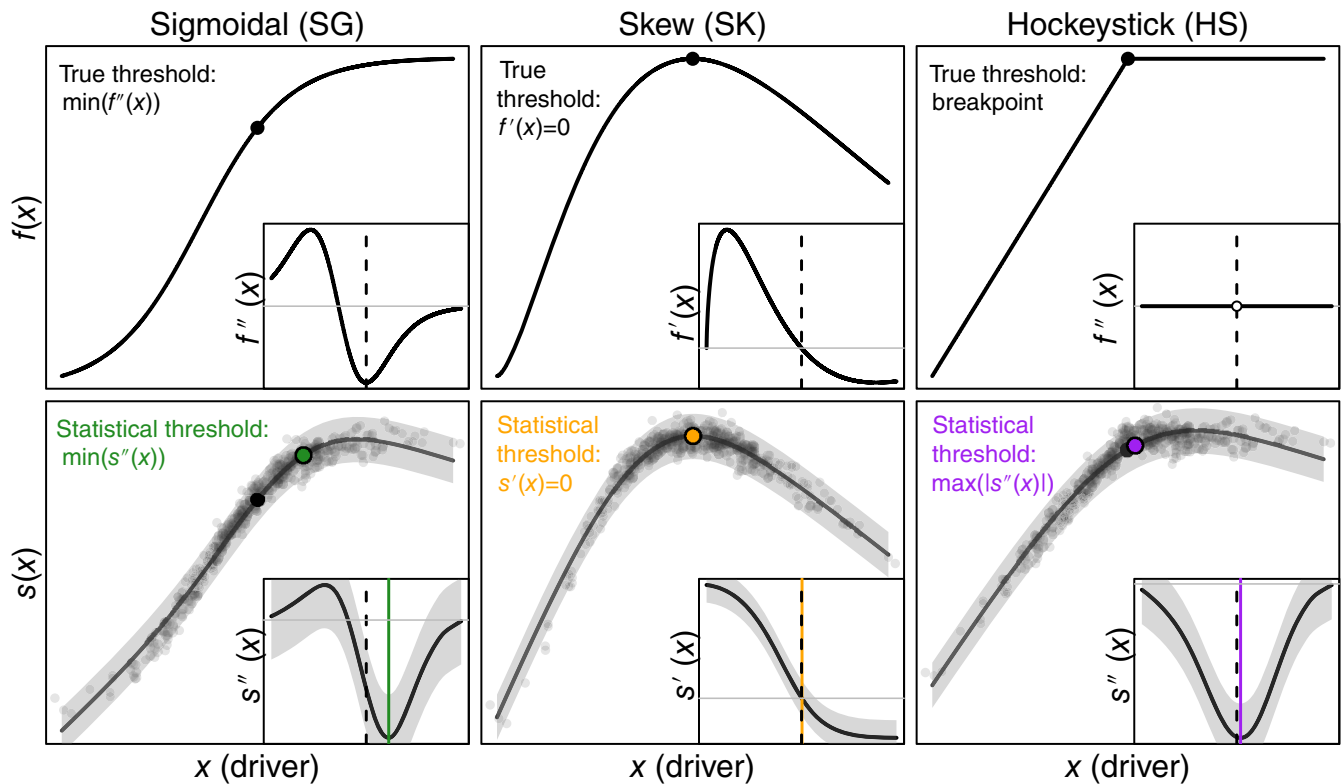


FIGURE 1 Suggested “true” (mathematical) and statistical (generalized additive model [GAM]-based) definitions of thresholds for the driver–response relationships used in the simulation analyses. The top row shows the true underlying relationship $f(x)$. The black points mark the true threshold, and the insets show the relevant first or second derivatives (with threshold location indicated by a dashed vertical line). The bottom row shows an example of hypothetical GAM fits $s(x)$ (black lines) to data (gray points) simulated from the true relationships and the derivatives of these fits (insets), with hypothetical CIs in gray. True thresholds are shown with black points (black dashed lines in insets), while statistically defined thresholds are shown in colored points (colored solid lines in insets). The gray horizontal lines in the derivative plots indicate where the value of the derivative is zero. See main text for justification of threshold definitions for each shape.

in the estimated driver–response relationship by analyzing the first and second derivatives of the GAM smoothing term $s(x)$ (Large et al., 2013; Samhoury et al., 2017). The first derivative (slope) of the smoothing term, $s'(x)$, provides information about local trends in $s(x)$ and locations of local optima (maxima or minima). The second derivative, $s''(x)$, allows for inferences about the rate at which these trends are changing. Large values of $s''(x)$ mean the slope of $s(x)$ is changing rapidly, indicating the response quickly shifts from being minimally to highly sensitive to changes in the driver (or vice versa). Derivatives were calculated using the derivatives function from the *gratia* package (Simpson, 2023). If the best-fitting model also included an additional covariate (which only occurred in the scenarios in which the additional covariate was assumed to be observed and thus included in the model selection process), we applied our threshold calculations to the marginal driver–response relationship instead. We considered four alternative statistical (i.e., GAM-based) threshold definitions:

1. The location where the absolute value of the second derivative is maximized ($\max(|s''(x)|)$), i.e., where the slope of $s(x)$ is changing most rapidly,
2. The location of the minimum in the second derivative with the greatest magnitude ($\min(s''(x))$), i.e., where the slope of $s(x)$ is decreasing most rapidly,
3. The location where the first derivative crosses zero ($s'(x) = 0$, i.e., a local optimum of $s(x)$), and
4. The location where the second derivative crosses zero ($s''(x) = 0$, i.e., a change in concavity of $s(x)$).

We explored these different definitions because the “threshold” location managers want to avoid may depend on the shape of the driver–response relationship as well as the context (Figure 1). For example, populations with compensatory growth exhibit a skew-shaped relationship between population size and per capita growth rate (Schreiber, 2003). When harvesting such populations, managers may be interested in finding the optimum in this relationship (i.e., the population size that produces

the maximum sustainable or economic yield) and/or the inflection point marking the boundary between relatively stable yield versus substantial suppression of productivity. Specifying different definitions also helps to address ambiguity in the literature related to the term “threshold” (Groffman et al., 2006). We explored the use of all four statistical definitions for each of our three threshold driver–response relationships, but in the main text we focused on the relationship/definition pairings shown in Figure 1.

Threshold robustness

We used jackknife (leave-one-out) resampling to evaluate the robustness of each detected threshold’s existence and location. We chose this approach both to reduce the computational costs of performing numerous simulation replicates and because jackknifing is more appropriate for our small sample sizes (time series lengths of 15–35 observations) than other resampling methods like bootstrapping. We differentiate the jackknife from simply generating new datasets in that the jackknife (or similar subsampling) allows one to estimate uncertainty for quantities of interest for a given dataset, while simulating new datasets allows one to explore variability across datasets. If a threshold was found in a sufficient fraction of jackknife iterations (we chose 0.25 as a default), we considered this support for the existence of that threshold and calculated the mean and 95% CIs of its estimated location across the jackknife iterations that detected it. We explored the use of the derivative’s simultaneous intervals (from the *gratia* package) and jackknifed derivative quantiles (upper and lower 95% quantiles of the GAM derivatives across all jackknife iterations) as alternative criteria to determine whether a threshold detected in an individual jackknife iteration was significant. More details on these calculations are provided in Appendix S1.

For each simulation scenario, we repeated our simulations 100 times to generate replicate datasets, calculated a mean threshold estimate for each using jackknife resampling, and used the resulting distribution of threshold estimates to characterize expected bias (difference from true threshold location) and precision (width of CIs around each estimate of threshold location). We also calculated true positive rates (TPRs; the fraction of simulation replicates that detected a threshold when one truly existed) and FPRs (the fraction of simulation replicates that detected a threshold when the true relationship was linear). When calculating TPRs, we counted any positive detection as a true positive, regardless of the accuracy of the estimated threshold location. The simulation and

threshold detection steps described above are summarized in Figure 2 and Appendix S1: Figures S1–S3.

Quantities of interest

We explored the sensitivity of GAM detection methods to four dimensions of interest, hereafter referred to as simulation parameters: (1) number of data points (which we refer to as time series length, but could also be spatial locations), (2) magnitude of variance in observation error, (3) distribution of the driver data relative to the threshold location, and (4) magnitude of the variance of the additional covariate. A priori, we expect thresholds to be more detectable with longer time series length, lower observation error variance, driver data centered around (i.e., on both sides of) the location of the threshold, and in the absence of any additional covariate effects. The parameters used in our simulations are summarized in Table 1. The range of time series lengths and observation errors we used are based on California Current Integrated Ecosystem Assessment (CCIEA) time series (Appendix S1: Figure S8). We chose these time series because they have been used in previous studies focused on detecting ecosystem thresholds and are publicly available (Samhuri et al., 2017).

Case study on Pacific hake

To demonstrate how our simulations can inform threshold detectability in empirical systems, we next investigated the potential for threshold relationships between annual temperature and the spatial distribution of Pacific hake (*M. productus*) along the west coast of North America. Based on the findings of Malick et al. (2020), we used the interpolated annual temperature anomaly at 100 m depth averaged between 45 and 49° N as our driver of interest. Malick et al. (2020) showed that warmer years in this region and depth were associated with more adult hake biomass in Canada, suggesting this temperature variable could represent an important driver of hake distribution. The data were collected as part of the acoustic trawl survey described in the next paragraph; see Malick et al. (2020) for details on data collection and calculation of interpolated temperature anomalies.

For the response variables, we calculated the proportion of hake biomass in Canada (north of 49° N) in summer using acoustic trawl survey data (see Berger et al., 2022; Fleischer et al., 2008; and Malick et al., 2020 for a detailed description of this survey). Because this survey continues north until hake are no longer detected, its

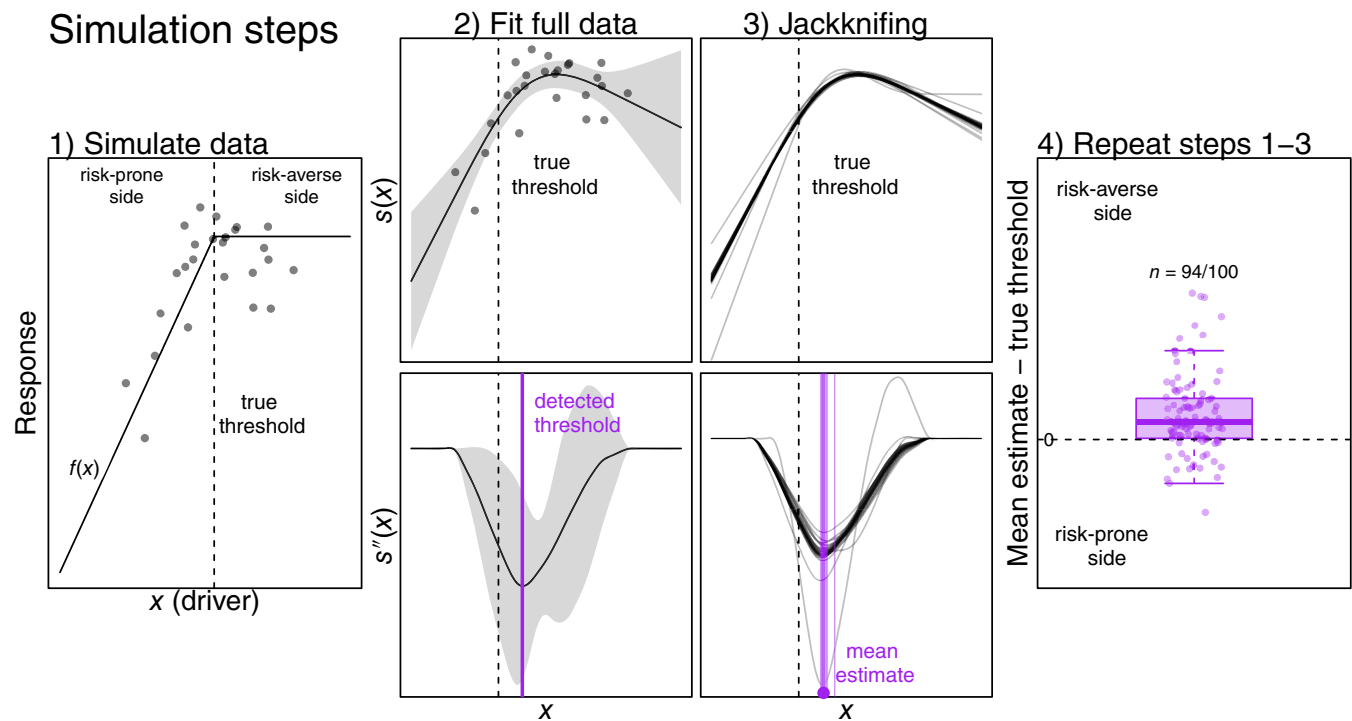


FIGURE 2 Summary of steps for using simulated data to test the ability of generalized additive models to detect thresholds in driver–response relationships (see main text for details). The hockey stick relationship is shown here as an example; see Appendix S1: Figures S1 and S2 for the other relationships. The dashed vertical line in the middle two columns indicates the true threshold location. For panels in the third column, each gray line shows the results from a single jackknife iteration; the vertical purple lines are the threshold estimates corresponding to each iteration, with the mean indicated by the purple point. The points in column 4 are the mean estimates from 100 simulated datasets. We assumed low values of the response variable to be less desirable, making threshold estimates to the right of the true threshold (preceding the abrupt decline in the response) “risk-averse” and estimates to the left “risk-prone.” False positive rates of our threshold detection methods were evaluated using a linear null model (Appendix S1: Figure S3).

TABLE 1 Simulation parameters and values used in simulations.

Parameter	Description	Values
Time series length	No. observations in simulated dataset	15, 25 ^a , 35
Observation error	SD of observation error in response variable	0.0275, 0.55 ^a , 2.75
Threshold quantile	Location of threshold relative to center of driver variable’s distribution	0.05, 0.25 ^a , 0.5
Covariate variability	SD of the (zero-centered) covariate	0 ^a , 0.5, 1

Note: The low, intermediate, and high values of observation error correspond to coefficients of variation of approximately 0.01, 0.2, and 1, respectively.

^aDefault values used in simulations.

northernmost extent varies year to year. To account for this sampling variability, we used only the data from transects within the common extent of the survey across all years. We considered both total biomass as well as

three age-specific groups (age 2 years, ages 3–4 years, and ages 5–20 years), as hake migration patterns and their relationships with ocean temperature vary by age (Agostini et al., 2008; Malick et al., 2020). This resulted in four driver–response relationships, to which we fit both GAMs and linear models. For relationships with evidence of nonlinearity, we calculated thresholds using the same procedure as for simulated datasets described earlier.

Because the hake time series were short (13 years of observations, roughly every other year from 1995 to 2022) and exhibited high variability, we used our simulation methods to evaluate effects of time series length and observation error on threshold detectability in simulated data with characteristics based on the hake data for ages 5–20 years. Briefly, we used the GAM fit to the full age 5–20 data to estimate the “true” driver–response relationship used in our simulations (approximated as a hockey stick relationship; Appendix S1: Figure S20). We ran simulations with time series lengths ranging from 13 (current) to 41 years in 4 year intervals with (1) current observation error (based on the SDs of the GAM and linear regression fits to the data) and (2) observation error

set to 10% of its current value (e.g., representing inclusion of additional covariates that account for some residual error—in which case the GAM fits can be thought of as the marginal predictions of the driver–response relationship). To evaluate FPRs, we repeated all simulations with a linear driver–response relationship, using the slope and intercept from a linear regression fit to the full data (Appendix S1: Figure S20). Due to the uncertainty in the true driver–response relationship, we used the $\max(|s''(x)|)$ threshold definition for all threshold calculations, as our simulation work suggested threshold estimates based on this definition were the most robust to the underlying driver–response relationship (Appendix S1: Figures S15 and S16). A full description of these simulations is provided in Appendix S1.

RESULTS

Simulation study

In general, longer time series, lower observation error, and threshold quantiles closer to the center of the driver's true distribution improved the likelihood of correctly detecting a threshold (i.e., increased the number of simulations with positive detection) and reduced bias in threshold estimates (Figure 3). For the parameter combinations we considered, observation error had the greatest effect on the likelihood of detecting a threshold, and time series length had the smallest effect on estimate bias (Figures 3 and 4). Estimates tended to shift from the data-rich side (assumed to be the risk-averse side in our analyses) toward the true threshold as threshold quantiles approached the center of the driver's distribution. When threshold quantile was equal to 0.5, estimates were more likely to exceed the true value, although the CIs around threshold estimates were rarely fully on the risk-prone side (Appendix S1: Figures S9 and S10).

Parameter effects differed slightly among the relationship/definition combinations. With the sigmoidal/ $\min(s''(x))$ combination, increasing time series length improved the likelihood of detecting a threshold but had minimal effects on estimate bias. Additionally, estimates of sigmoidal/ $\min(s''(x))$ thresholds tended to be more biased than for the other two relationship/definition combinations, and this bias was mainly toward the risk-averse side. Across all parameter combinations, threshold detectability was highest for the hockey stick/ $\max(|s''(x)|)$ combination and estimates were generally most precise for the skew/ $s'(x) = 0$ combination (Figures 3 and 4; Appendix S1: Figures S9 and S10).

FPRs were generally low ($< \sim 10\%$) across the parameter combinations we considered and were always lower

than corresponding TPRs (Figure 4; Appendix S1: Figure S11). FPRs tended to be highest when using the $\max(|s''(x)|)$ and $s''(x) = 0$ threshold definitions and lowest with the $s'(x) = 0$ definition. Simulation parameters had stronger effects on TPRs than FPRs. In general, TPRs were highest for long time series, low observation error, and threshold quantiles closer to the center of the driver's distribution, although sigmoidal/ $\min(s''(x))$ and hockey stick/ $\max(|s''(x)|)$ TPRs were highest when the threshold quantile was equal to 0.25 (Figure 4).

When the SD of the additional covariate was nonzero, ignoring this covariate in the model fitting process resulted in lower TPRs and more biased threshold estimates than when covariate variability was equal to zero. These effects were greater for higher levels of covariate variability, particularly when the additional covariate had a nonlinear effect on the response (Figure 5; Appendix S1: Figure S12). Including the additional covariate in candidate models increased TPRs, reduced estimate bias, and generally reduced FPRs (Figure 5; Appendix S1: Figure S12).

The above results were for the relationship/threshold definition pairings shown in Figure 1. To confirm our proposed pairings were reasonable, we also considered all possible combinations of our four threshold definitions and three focal relationships, as well as several “nonfocal” driver–response relationships (Appendix S1: Figures S13–S16). For all focal relationships, the $\max(|s''(x)|)$ and $\min(s''(x))$ definitions produced identical estimates (as the minimum value of $s''(x)$ was the point with the greatest absolute value), while the $s''(x) = 0$ estimates were the most variable and biased (Appendix S1: Figure S15). Under most conditions, the $\max(|s''(x)|)$ and $\min(s''(x))$ definitions performed fairly well for all three relationships, whereas the $s'(x) = 0$ was only a good choice for the skew relationship (Appendix S1: Figure S15). These patterns generally held for the nonfocal shapes we explored, although the $\max(|s''(x)|)$ and $\min(s''(x))$ definitions were no longer the same for the nonfocal skew and hockey stick relationships (with the $\max(|s''(x)|)$ definition being more accurate; Appendix S1: Figure S16).

Adding significance criteria to threshold estimates had overall minimal effects on estimate bias, at least at our default parameter values (Appendix S1: Figure S17). Using jackknifed quantiles to evaluate significance generally did not alter TPRs or FPRs, while using simultaneous intervals reduced both (Appendix S1: Figures S18 and S19). The exception to this was the $s''(x) = 0$ definition, for which FPRs remained relatively unchanged with the addition of either significance criterion (Appendix S1: Figure S18). Increasing the minimum

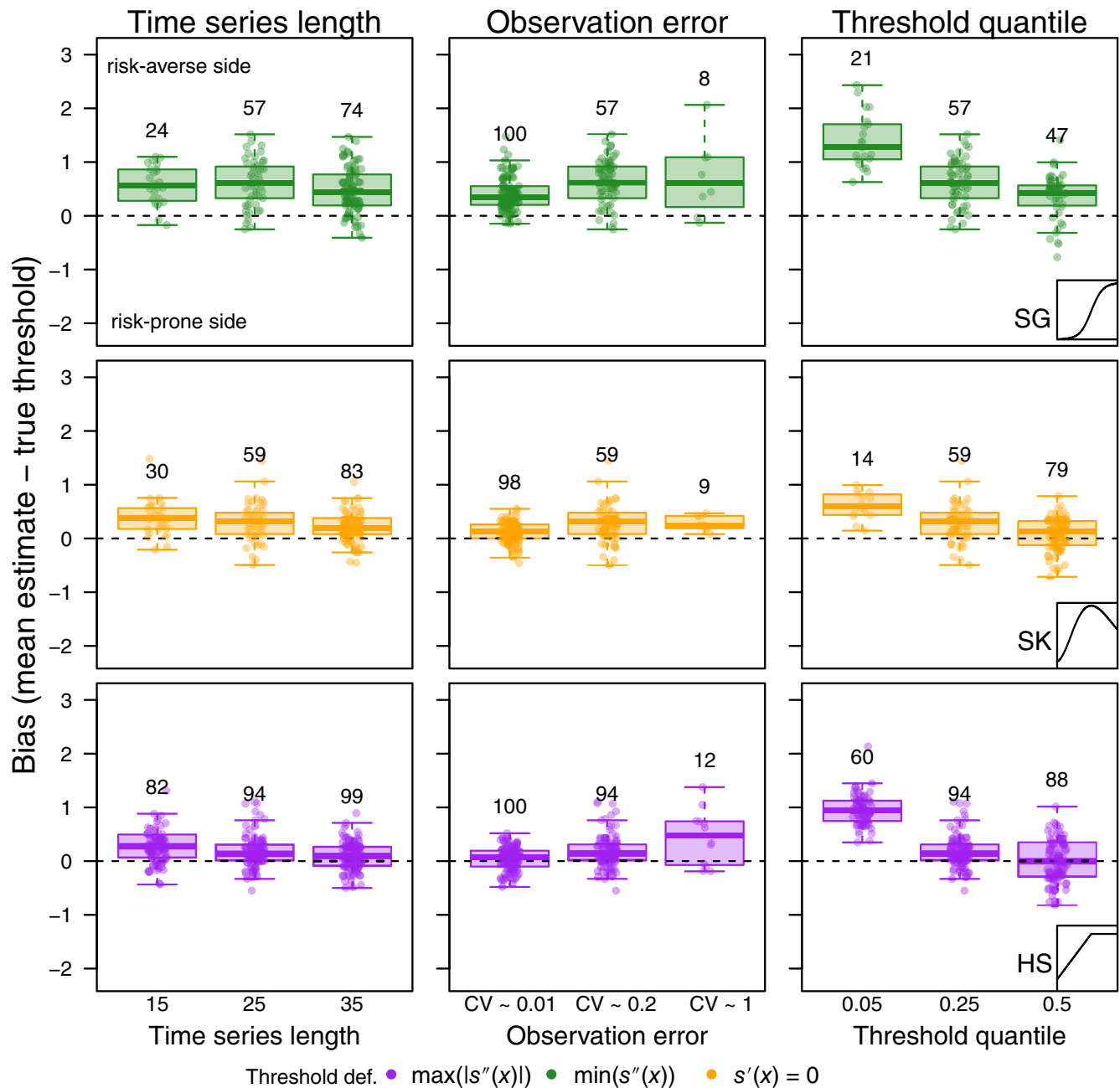


FIGURE 3 Effects of simulation parameters on threshold detectability. The x -axis is the length of the simulated time series (left column), the approximate CV in the response variable (middle column), or the quantile of the driver in which the true threshold occurs (right column). Parameters not being varied were set to their default values; that is, time series length = 25, observation error = 0.55 (resulting in a response CV of ~ 0.2), and threshold quantile = 0.25. The y -axis is the difference between a simulation's mean threshold estimate across jackknife iterations and the true threshold, in units of SDs of the driver. Positive values indicate the estimated threshold was on the "risk-averse" side, while negative values indicate the estimated threshold was on the "risk-prone" side. Sample sizes above the boxplots represent the number of simulations that detected a threshold, out of a total of 100 replicates. The top, middle, and bottom rows show results for sigmoidal (SG), skew (SK), and hockey stick (HS) driver-response relationships, respectively. Colors indicate the default definition used for calculating the threshold for each relationship.

fraction of jackknife iterations required to detect a threshold similarly had no effect or decreased both TPRs and FPRs (Appendix S1: Figure S19). Given that stricter criteria for threshold significance either had

minimal effects or reduced both TPRs and FPRs, these additional criteria may be most important when FPRs are of great concern and the reduction in TPRs is an acceptable trade-off.

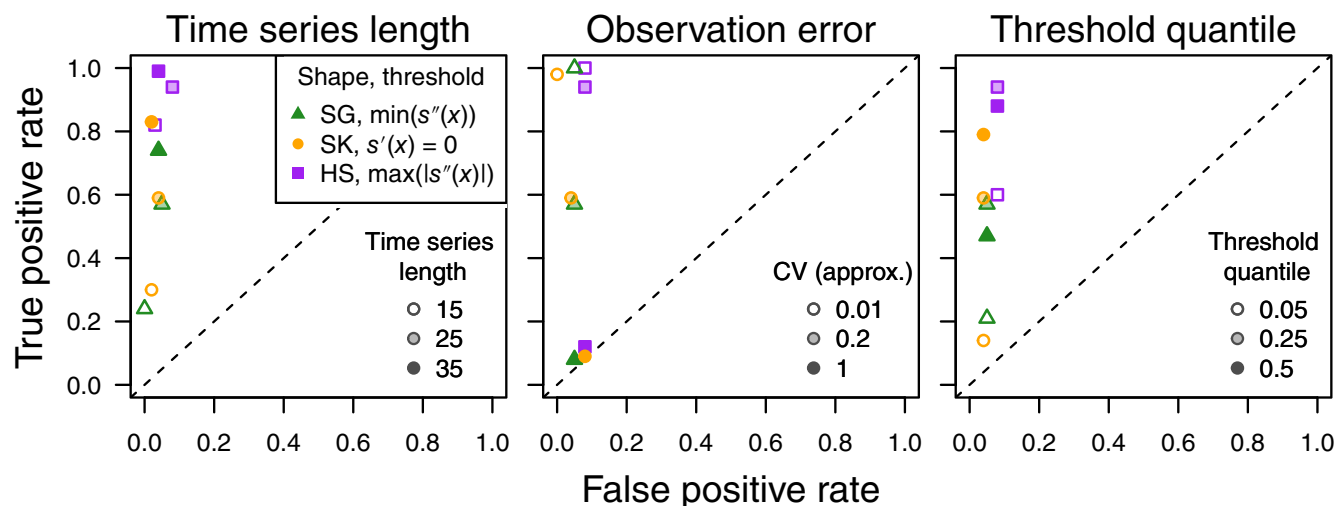


FIGURE 4 Receiver-operator curves showing effects of simulation parameters on threshold detectability. The x-axis is the fraction of simulations (out of 100) that detected a threshold when the true relationship was linear. The y-axis is the fraction of simulations that detected a threshold (regardless of the accuracy of the threshold estimate) when the true relationship was sigmoidal (SG), skew (SK), or hockey stick (HS), as indicated by the shape and color of the points. The effects of time series length, observation error (approximate CV), and threshold quantile are shown in the left, middle, and right columns, respectively, with symbol shading indicating the value of each parameter. Parameters not being varied were held at their default values. The dashed line is the 1:1 line, representing a random guess.

Hake case study

Analysis of the hake acoustic survey data revealed no support for a nonlinear relationship between 45 and 49° N temperature anomalies and the proportion of total and age 5–20 hake biomass in Canada ($\Delta\text{AIC}_c \sim 1.6$ in favor of the linear model) (Figure 6). There was also no evidence of nonlinear temperature effects on any of the other age groups (Appendix S1: Figure S21). For the age 5–20 data, we detected a threshold in the full dataset and in 9 out of 13 leave-one-out jackknife iterations without applying any significance criteria. However, when applying the jackknifed quantiles and simultaneous interval significance criteria, significant thresholds were only detected in 4/13 and 2/13 iterations, respectively, indicating this threshold is not significant.

Although there was no AIC support for the nonlinearity in the age 5–20 group, the small sample size and high level of noise in the data makes it difficult to make definitive conclusions. Our simulations provided greater insight into the level of confidence we should expect to have in the lack of a significant nonlinear relationship. For conditions based on the current age 5–20 dataset and temperatures from the focal region of the US West Coast, the simulations suggest the likelihood of detecting a true threshold in the relationship between temperature anomaly and proportion of age 5–20 hake biomass in Canada is low and approximately equal to the likelihood of incorrectly detecting a threshold if this relationship is linear (Figure 7). When thresholds were

detected in these simulations, their estimated locations were variable and often highly biased (Appendix S1: Figure S22). Increasing time series length generally resulted in higher TPRs but did not consistently affect FPRs or reduce variability in locations of estimated thresholds (Figure 7; Appendix S1: Figure S22). For a given time series length, reducing observation error greatly increased TPRs and reduced estimate bias but had minimal effects on FPRs (Figure 7; Appendix S1: Figures S22–S24). A reduction in FPRs was achieved by a combination of adding significance criteria (simultaneous intervals) to jackknifed thresholds and increasing the fraction of jackknife iterations required to detect a threshold, but these changes also reduced TPRs—particularly when time series were short (Figure 7).

DISCUSSION

Simulation study

Depending on the characteristics of the data, we found that GAMs can be a robust method for detecting thresholds in simple driver–response relationships. The levels of bias and TPRs/FPRs acceptable to managers will be context-dependent and influenced by the relative risks of missing versus incorrectly detecting a threshold. However, our results offer some general guidelines, summarized in Tables 2 and 3. For parameter combinations representative of real systems, we found that observation

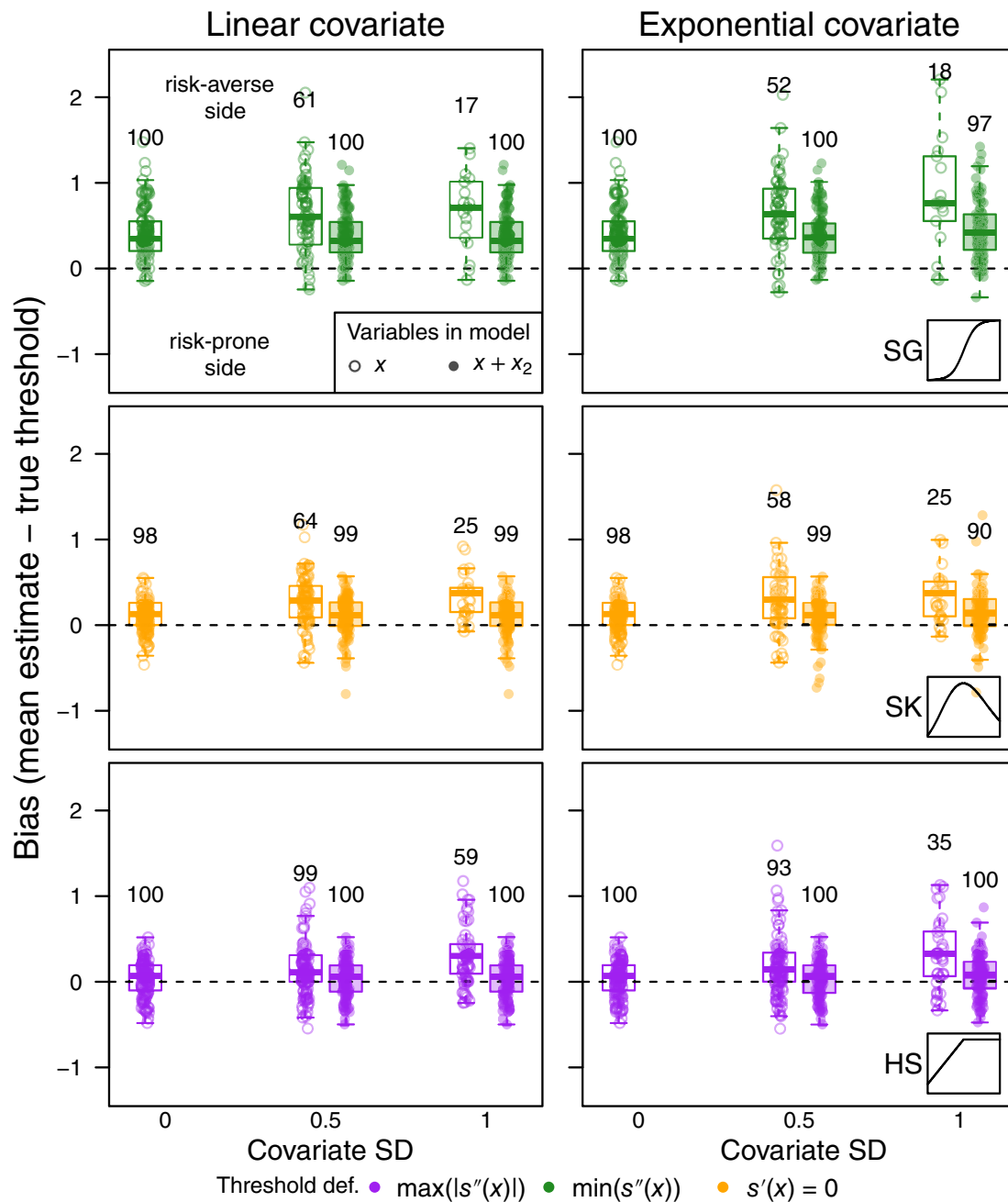


FIGURE 5 Effect of a missing covariate on threshold detectability. Values of this additional covariate were drawn from a normal distribution with a mean of 0 and a SD of 0 (no covariate effect), 0.5, or 1, indicated on the x-axis. The relationship between the additional covariate and the response was either linear or exponential. Simulated data were fit with generalized additive models that included only the driver x as an explanatory variable (open circles/boxes) and again with models that included both the driver and additional covariate x_2 (solid circles/boxes). The y-axis shows the resulting difference between each simulation's mean threshold estimate and the true threshold, in units of SDs of the driver. Numbers indicate the number of simulations (out of 100) that detected a threshold. The top, middle, and bottom rows show results for sigmoidal (SG), skew (SK), and hockey stick (HS) driver-response relationships, respectively. Colors indicate the default definition used for calculating the threshold for each relationship. Other parameters: time series length = 25, threshold quantile = 0.25, observation error = 0.0275 (CV of ~0.01).

error and threshold quantile tended to have stronger effects on threshold detection than time series length. Observation error can be easily estimated for a given dataset, providing a baseline for the expected accuracy of

GAM-based threshold detection. If a time series has a high CV (e.g., of one or more), our findings indicate it may not be possible to draw robust conclusions about the existence of a threshold with the detection methods used here.

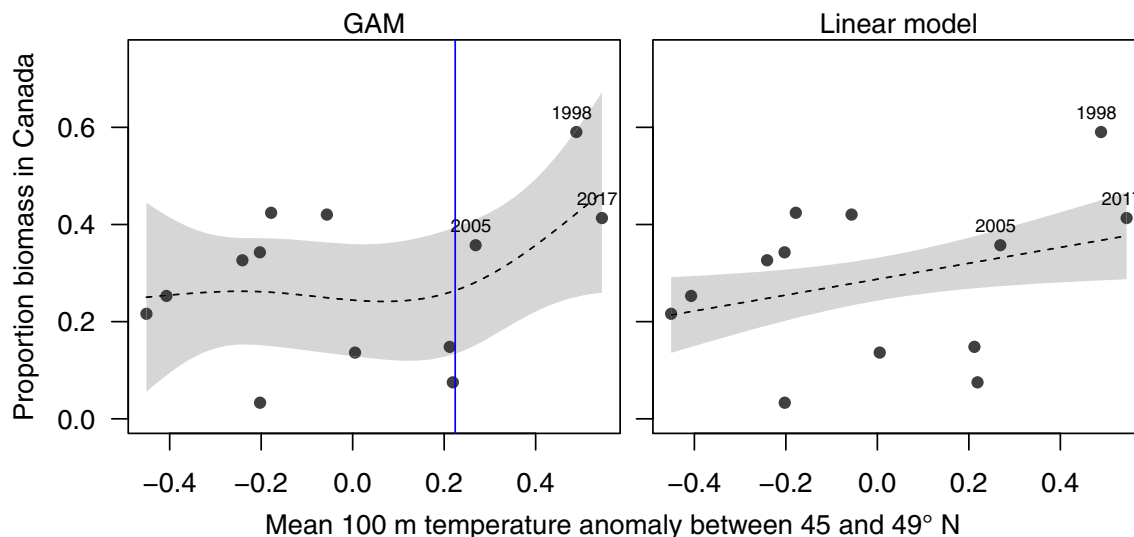


FIGURE 6 Results of model fitting and threshold detection applied to the age 5–20 hake data. The x-axis is the driver (mean interpolated temperature anomaly at 100 m depth) and the y-axis is the response (proportion age 5–20 hake biomass in Canada). The left panel shows the mean and simultaneous intervals of the generalized additive model (GAM) predictions, while the right panel shows the mean and SE of the linear regression predictions (neither of these fits were statistically significant). The blue line is the mean location of the nine thresholds detected across jackknife iterations (total iterations = 13) when no significance criteria were applied; however, this threshold was not significant (see main text). Extreme points are labeled with their corresponding years.

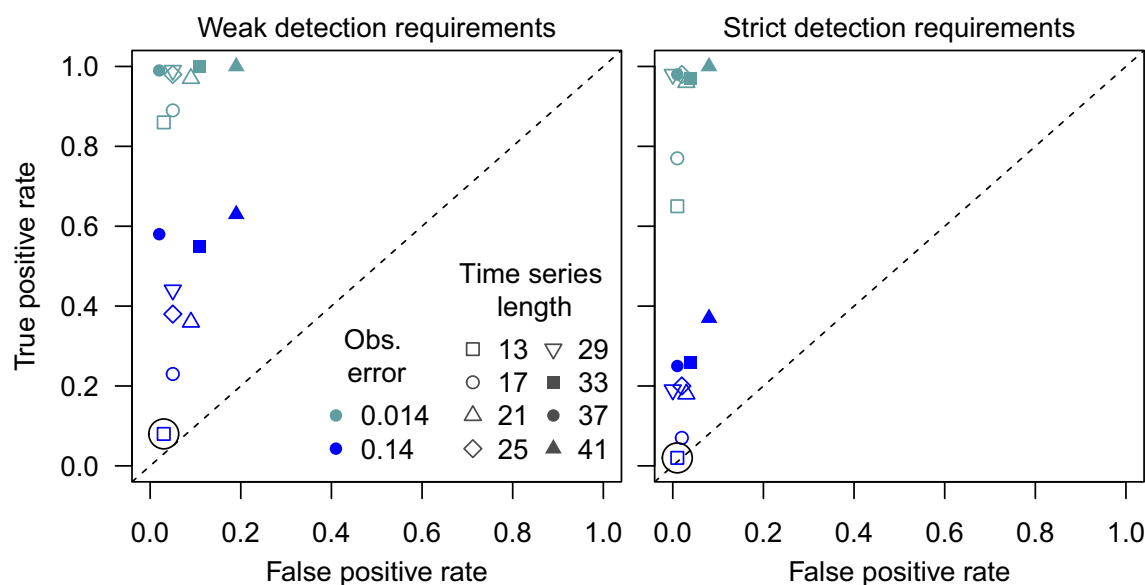


FIGURE 7 Receiver-operator curve showing effects of time series length (symbols) and observation error (colors, with corresponding values representing the SDs of the observation error) on false and true positive rates associated with applying generalized additive model threshold detection definitions to simulated temperature-hake data. False positive rate is the probability of detecting a threshold if the true relationship between temperature and proportion of age 5–20 hake biomass in Canada is linear, while true positive rate is the probability of detecting a threshold if the true relationship has a hockey stick shape. The dashed line is the 1:1 line. Conditions that approximate the current age 5–20 hake biomass and temperature dataset are circled in black. Left panel: No significance criteria and only 25% of jackknife iterations are required to detect a threshold for a positive detection. Right panel: Simultaneous intervals are used as significance criteria and at least 95% of jackknife iterations are required to detect a threshold for positive detection.

Threshold quantile is more challenging to infer from data than observation error. However, our simulations suggest that when observations are strongly skewed

toward one side of the threshold, threshold estimates tend to be biased toward the side with more data (the “risk-averse” side)—highlighting the need to evaluate

TABLE 2 Summary of general findings from our simulation study (TPR = true positive rate, FPR = false positive rate).

Data characteristics	
Simulation parameter	Effect on threshold detectability*
Time series length	Moderate; main effect of increasing time series length was to increase TPRs.
Observation error	Strong; detection not reliable for CV of ~1 or more (TPRs low and approx. equal to FPRs).
Threshold quantile	Strong; value of 0.05 (threshold exceeded in only ~5% of observations) produced low TPRs and large bias towards data-rich side of threshold.
Covariate variability	Can be strong, but including additional covariate in the estimation model reduces effects.

*Over the range of values considered in this study.

potential bias in tipping points a priori, and to be cautious when drawing conclusions based on a small number of extreme data points (as discussed in Satterthwaite et al., 2020). The direction of bias (toward the risk-averse side) was generally consistent across the relationship/definition combinations we explored. This finding suggests the GAM detection method comes with a relatively low risk of mistakenly concluding a threshold lies far on the undesirable side of its true value (making it unlikely the true threshold will be unexpectedly crossed before its estimated value is reached). However, the counter effect of this bias toward “risk-averse” estimates is that one might anticipate a tipping point earlier than it would manifest. If this conservative bias is not accounted for in communication and advice, failure of the tipping point to manifest at the (biased) anticipated point could undermine confidence in the existence of that tipping point, reducing management motivation for strong action (e.g., rapid scaling back of pressure) at the critical approach to the true tipping point. Finally, we focused on the frequency of threshold crossing, but a deeper exploration of extreme points should also consider the effects of how far past the threshold the data points lie, as this might affect how accurately we can capture the shape of the threshold relationship.

Another key takeaway from our simulations was that threshold detectability varies among relationship/definition combinations. For our three focal combinations, TPRs and FPRs tended to be highest for the hockey stick/ $\max(|s''(x)|)$ combination, and estimates were generally most biased for the sigmoidal/ $\min(s''(x))$ combination. We suggested threshold definitions we felt

TABLE 3 Summary of main recommendations from our simulation study.

Choice of threshold definition	
Underlying driver-response relationship	Recommended statistical threshold definition
Hockeystick	$\max(s''(x))$
Sigmoidal	$\max(s''(x))$ or $\min(s''(x))$
Skew	$\max(s''(x))$ or $s'(x) = 0$

made the most sense for specific driver–response relationships, but if the true driver–response relationship is uncertain, then our simulations indicate the $\max(|s''(x)|)$ definition (which is similar to definitions used by previous studies; Large et al., 2013; Samhour et al., 2017) would be the best choice and the $s''(x) = 0$ (inflection point) definition would be the most error prone. If using our focal relationship/definition combinations, this means mistakenly concluding a skew relationship is hockey stick and using the $\max(|s''(x)|)$ threshold definition would lead to better estimates than mistaking a hockey stick relationship for skew and using the $s'(x) = 0$ definition.

High levels of noise in the simulated data strongly decreased threshold detectability. However, when this noise was at least partially due to a missing covariate, we found that threshold detectability could be improved by including this covariate in the estimation model and applying threshold detection methods to the marginal predictions of the driver–response relationship. Thus, when driver–response relationships are noisy, these results suggest it may be helpful to include suspected covariates in estimation models and use model selection to choose a best-fitting model before applying threshold detection techniques. For example, with species distribution models, this might mean that a threshold effect of temperature on species distributions could be more easily detected if known covariates (e.g., variables related to habitat suitability; McHenry et al., 2019) are included in the model as well. However, it is important to note that our analyses assumed the covariate–response relationship was constant over time, and in real systems, nonstationarity could reduce the potential for covariate inclusion to reduce noise in the response variable (see *Future directions*).

The approaches developed in this simulation study are widely applicable across taxa and systems and may be particularly useful in the context of ecosystem-based fishery management. For example, in the United States, there is strong interest in advancing the integration of climate and ecosystem information in harvest-setting processes for federal fisheries (Schmidt et al., 2019), such as

through direct integration of environmental indices into stock assessments (e.g., Haltuch et al., 2019), development of risk tables to inform recommendations around acceptable biological catch (Dorn & Zador, 2020), and creation of ecosystem and socioeconomic profiles to provide context for harvest decisions (Shotwell et al., 2023). In all cases, environmental drivers form a basis for modifying insights from stock assessments. The threshold detection method we evaluated can provide a useful approach for translating environmental drivers into stoplight warning systems for decisions around harvest (Caddy, 2015) by providing objective, nonarbitrary cut-offs between favorable and unfavorable values of the drivers.

Hake case study

Our case study on Pacific hake serves as an example of how the simulation methods developed in this paper can be applied to real (and noisy) empirical data. Although we did not find evidence for any statistically significant relationship, the results of our analyses are qualitatively consistent with Malick et al.'s (2020) findings that higher temperatures in some regions may correspond to more adult hake in Canada. Our simulations suggest these findings are inconclusive due to the nature of the data (i.e., short and noisy time series), with low likelihoods of true as well as false threshold detection.

The results of our simulations also indicated that a combination of identifying covariates that are contributing to observed noise and increasing time series length could greatly improve threshold detectability in the Pacific hake system (Appendix S1: Figure S22). There are several covariates that may affect the proportion of adult hake that migrate into Canada, including the California Undercurrent (Agostini et al., 2006, 2008; Malick et al., 2024), the spatial distribution of euphausiids or other high-quality prey (Phillips et al., 2022, 2023), and dissolved oxygen (particularly off Vancouver Island; R. Thomas, C. Stanley, and S. Gauthier, personal communication). Future work could explore whether inclusion of these or other covariates reveals support for a threshold relationship with temperature (and/or a threshold effect of the covariates themselves). Another next step could be to apply our simulation approach to longer time series, such as datasets that sample young of the year (yoy) and larval hake in the California Current Ecosystem (Gallo et al., 2022). For example, the California Cooperative Oceanic Fisheries Investigation (CalCOFI) program has sampled larval hake (proxy for spawning stock biomass) at least quarterly since 1951 between the US/Mexico border and San Francisco (McClatchie, 2014). The simulation

methods developed in the present study could be used to explore the potential for threshold relationships between environmental variables and hake recruitment, which could provide greater insight into environmental effects on this important fishery species.

The basic approach used in the hake case study (running simulations with underlying driver–response relationships based on linear regression and GAM fits to the observed data) could easily be applied to any driver–response data (see Detmer & Ward, 2024 for our code). However, there are several assumptions to keep in mind when interpreting the results. Given the short time series in the hake dataset, we could not determine the distribution of the driver and for simplicity assumed it to be normal. Furthermore, we ignored nonstationarity, which would likely affect threshold detectability if present (see *Future directions*). We assumed the “true” nonlinear relationship could be approximated by the GAM fit to the observed data, which may not be the case. One could address this by repeating the simulations using a suite of potential driver–response relationships; however, our theoretical simulations suggest the effects of time series length and observation error are consistent across driver–response relationships, so misspecification of the true relationship is unlikely to affect our main conclusions here. Finally, we represented the effects of adding missing covariates as lowering observation error. Reducing noise in the empirical data by instead adding an additional covariate might give slightly different results, as threshold calculations would then be done on predicted marginal driver–response relationships. However, our simulation studies suggest adding a missing covariate and reducing observation error have similar effects on threshold detectability (Figure 5 vs. Figure 3), so lowering observation error should reasonably approximate the effects of including (noninteractive) covariates in the estimation model.

Future directions

This study represents an initial exploration of GAMs as a tool to detect ecological thresholds in driver–response relationships, providing a baseline for future studies. We focused on the effects of data characteristics commonly encountered in ecological research: time series length, observation error, frequency of threshold crossing, and a (potentially) latent additional covariate. However, a number of other factors could influence the robustness of threshold detection, including temporal and/or spatial autocorrelation (in which case either generalized additive mixed models or generalized linear mixed models with breakpoint functions would be more appropriate estimation models than GAMs; Large et al., 2013; Essington

et al., 2022; Samhouri et al., 2017), nonstationarity (e.g., shifts in the driver–response relationship and/or driver distribution over time; Chavez-Demoulin & Davison, 2005; Puerta et al., 2016; Recknagel et al., 2013), and more complex driver–response–covariate relationships. The latter could include driver–response relationships with multiple thresholds (such as relationships that exhibit hysteresis, which require more advanced modeling techniques to detect; Andersen et al., 2009) and interactions between one or more covariates and the driver (potentially creating multivariate thresholds and/or thresholds emerging from an interaction between linear drivers and covariates; Crain et al., 2008). The basic simulation methods we used here for GAMs could be adapted to explore the robustness of more complex models better suited to these types of scenarios, which are likely more reflective of real systems.

Future work could also explore additional methods for testing GAM robustness. For example, a key step in our simulations was to determine whether a driver–response relationship was nonlinear. Previous empirical threshold detection studies have used a variety of methods to determine whether nonlinearities are significant, including comparison of the AIC scores of linear and nonlinear models, estimated degrees of freedom (EDF), the significance of the smoother term, and the effect of the smoother term on the model's GCV score (Large et al., 2013; Samhouri et al., 2017; Satterthwaite et al., 2020). For simplicity, we focused here on the AIC score, but we acknowledge that there are challenges associated with AIC-based model selection (Sutherland et al., 2023). These include the arbitrary $\Delta\text{AIC} > 2$ requirement as well as issues with using AIC selection to draw conclusions about error rates (complexity in AIC is a function of parameters, and with 1 degree of freedom the significance level is fixed at 0.157 rather than 0.05; Anderson & Burnham, 2002; Forster, 2000; Kuha, 2004). Moving beyond traditional AIC-based model selection for threshold detection, it may be worthwhile to explore these questions in a Bayesian framework; extensions include examining the utility of penalties or priors (Hooten & Hobbs, 2015), using log scores to incorporate both accuracy and precision (Krnjajić & Draper, 2014), or implementing multi-model ensembling.

Finally, our analyses focused on the ability of GAMs to detect nonlinear, threshold relationships. We therefore defined a false positive as detecting a threshold when the underlying relationship was linear. However, another type of false positive is the spurious detection of a threshold when no relationship (linear or nonlinear) exists at all. Similarly, a false negative could be defined as failing to detect any relationship when the driver does in fact influence the response. Depending on the scenario, these

alternative definitions may be of greater interest to managers. For example, if the true relationship was linear, the best management decision would likely be a continuous or proportionate response to variation in the driver, but it might still be better to spuriously detect a threshold and enact a stepped response to extreme values of the driver than to miss the relationship completely and have no response at all. Incorporating these alternative definitions of true and false positives into the simulation work we introduced here could provide managers with more nuanced insight into management decisions involving GAM-based threshold detection.

CONCLUSION

This study provides an initial assessment of the ability of GAMs to detect thresholds in simple bivariate relationships, highlighting how data characteristics (particularly observation error and threshold quantile) and choice of threshold shape/definition pairing can have strong effects on detectability. Overall, we found GAMs to be a fairly robust detection method, although more work is needed to investigate how they perform under more complex (and realistic) scenarios. One particular challenge when working with real ecological data is that they are often quite noisy, which can make it difficult to draw firm conclusions about the existence of thresholds (Large et al., 2013; Poikane et al., 2022; Samhouri et al., 2017). We chose a case study with ambiguous results to illustrate how simulations can provide additional insight in these situations (e.g., by suggesting what sample sizes are needed for robust threshold estimates given the level of noise in the data). As ecosystems around the world become increasingly stressed by anthropogenic pressures and growing environmental variability, analyses like ours can help to guide the implementation of ecosystem-based management strategies that are built around an understanding of thresholds.

AUTHOR CONTRIBUTIONS

All authors were involved in conceptualization of the study. A. Raine Detmer, Mary E. Hunsicker, Jameal F. Samhouri, and Eric J. Ward designed the research, and A. Raine Detmer conducted the analyses. A. Raine Detmer, Mary E. Hunsicker, Jameal F. Samhouri, and Eric J. Ward drafted the manuscript. All authors contributed to writing the final draft.

ACKNOWLEDGMENTS

We thank Rebecca Thomas, Chelsea Stanley, and Stéphane Gauthier for helpful discussions about drivers of hake distributions. We thank Sarah Gaichas, Chris

Harvey, Corey Ridings, and Kate Richerson for their participation in the NOAA thresholds workshop that inspired this study. We thank the NOAA Integrated Ecosystem Assessment program and the ICES/PICES Strategic Initiative for Climate Change Impacts on Marine Ecosystems for sharing threshold detection R code. We thank the NOAA Integrated Ecosystem Assessment Program for funding Jameal F. Samhour and Elliott L. Hazen salary. Jameal F. Samhour and Kristin N. Marshall thank TS for inspiration. This research was supported by an NSF INTERN to A. Raine Detmer.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data and code (Detmer & Ward, 2024) are available from Zenodo: [10.5281/zenodo.14210416](https://doi.org/10.5281/zenodo.14210416).

ORCID

A. Raine Detmer  <https://orcid.org/0000-0001-5416-0278>

Eric J. Ward  <https://orcid.org/0000-0002-4359-0296>

Kelly S. Andrews  <https://orcid.org/0000-0001-7734-3800>

Bridget E. Ferriss  <https://orcid.org/0000-0002-6739-5313>

Elliott L. Hazen  <https://orcid.org/0000-0002-0412-7178>

Michael Malick  <https://orcid.org/0000-0002-8376-5476>

Kiva L. Oken  <https://orcid.org/0000-0002-5627-6837>

REFERENCES

- Agostini, V. N., R. C. Francis, A. B. Hollowed, S. D. Pierce, C. Wilson, and A. N. Hendrix. 2006. "The Relationship between Pacific Hake (*Merluccius productus*) Distribution and Poleward Subsurface Flow in the California Current System." *Canadian Journal of Fisheries and Aquatic Sciences* 63: 2648–59.
- Agostini, V. N., A. N. Hendrix, A. B. Hollowed, C. D. Wilson, S. D. Pierce, and R. C. Francis. 2008. "Climate–Ocean Variability and Pacific Hake: A Geostatistical Modeling Approach." *Journal of Marine Systems* 71: 237–248.
- Aminikhanghahi, S., and D. J. Cook. 2017. "A Survey of Methods for Time Series Change Point Detection." *Knowledge and Information Systems* 51: 339–367.
- Andersen, T., J. Carstensen, E. Hernández-García, and C. M. Duarte. 2009. "Ecological Thresholds and Regime Shifts: Approaches to Identification." *Trends in Ecology & Evolution* 24: 49–57.
- Anderson, D. R., and K. P. Burnham. 2002. "Avoiding Pitfalls when Using Information-Theoretic Methods." *The Journal of Wildlife Management* 66: 912.
- Benedetti, A., M. Abrahamowicz, K. Leffondré, M. S. Goldberg, and R. Tamblyn. 2009. "Using Generalized Additive Models to Detect and Estimate Threshold Associations." *The International Journal of Biostatistics* 5(1): 26.
- Berger, A., A. Edwards, C. Grandin, and K. Johnson. 2022. "Status of the Pacific Hake (Whiting) Stock in U.S. and Canadian Waters in 2022." Prepared by the Joint Technical Committee of the U.S. and Canada Pacific Hake/Whiting Agreement. National Marine Fisheries Service and Fisheries and Oceans Canada. 238 p.
- Caddy, J. F. 2015. "The Traffic Light Procedure for Decision Making: Its Rapid Extension from Fisheries to Other Sectors of the Economy." *Global Journal of Science Frontier Research: I Marine Science* 15(1): 30.
- Carpenter, S. R., W. A. Brock, J. J. Cole, and M. L. Pace. 2014. "A New Approach for Rapid Detection of Nearby Thresholds in Ecosystem Time Series." *Oikos* 123: 290–97.
- Chavez-Demoulin, V., and A. C. Davison. 2005. "Generalized Additive Modelling of Sample Extremes." *Journal of the Royal Statistical Society: Series C: Applied Statistics* 54: 207–222.
- Crain, C. M., K. Kroeker, and B. S. Halpern. 2008. "Interactive and Cumulative Effects of Multiple Human Stressors in Marine Systems." *Ecology Letters* 11: 1304–15.
- Daily, J. P., N. P. Hitt, D. R. Smith, and C. D. Snyder. 2012. "Experimental and Environmental Factors Affect Spurious Detection of Ecological Thresholds." *Ecology* 93: 17–23.
- Detmer, A. R., and E. Ward. 2024. "nwfs-cb/Threshold-Detection: v1.0.0 [Data and Code]." Zenodo. <https://doi.org/10.5281/zenodo.14210416>.
- Deutsch, C. A., J. J. Tewksbury, R. B. Huey, K. S. Sheldon, C. K. Ghalambor, D. C. Haak, and P. R. Martin. 2008. "Impacts of Climate Warming on Terrestrial Ectotherms across Latitude." *Proceedings of the National Academy of Sciences of the United States of America* 105: 6668–72.
- Dorn, M. W., and S. G. Zador. 2020. "A Risk Table to Address Concerns External to Stock Assessments when Developing Fisheries Harvest Recommendations." *Ecosystem Health and Sustainability* 6: 1813634.
- Essington, T. E., S. C. Anderson, L. A. K. Barnett, H. M. Berger, S. A. Siedlecki, and E. J. Ward. 2022. "Advancing Statistical Models to Reveal the Effect of Dissolved Oxygen on the Spatial Distribution of Marine Taxa Using Thresholds and a Physiologically Based Index." *Ecography* 2022: e06249.
- Ficetola, G., and M. Denoël. 2009. "Ecological Thresholds: An Assessment of Methods to Identify Abrupt Changes in Species–Habitat Relationships." *Ecography* 32: 1075–84.
- Fleischer, G. W., K. D. Cooke, P. H. Ressler, R. E. Thomas, S. K. de Blois, and L. C. Hufnagle. 2008. "The 2005 Integrated Acoustic and Trawl Survey of Pacific Hake, *Merluccius productus*, in U. S. and Canadian Waters off the Pacific Coast." U.S. Department of Commerce, NOAA Technical Memorandum NMFS-NWFSC-94: 41.
- Forster, M. R. 2000. "Key Concepts in Model Selection: Performance and Generalizability." *Journal of Mathematical Psychology* 44: 205–231.
- Gallo, N. D., N. M. Bowlin, A. R. Thompson, E. V. Satterthwaite, B. Brady, and B. X. Semmens. 2022. "Fisheries Surveys Are Essential Ocean Observing Programs in a Time of Global Change: A Synthesis of Oceanographic and Ecological Data from U.S. West Coast Fisheries Surveys." *Frontiers in Marine Science* 9: 757124.
- Gove, J. M., G. J. Williams, J. Lecky, E. Brown, E. Conklin, C. Counsell, G. Davis, et al. 2023. "Coral Reefs Benefit from Reduced Land–Sea Impacts under Ocean Warming." *Nature* 621: 536–542.

- Groffman, P. M., J. S. Baron, T. Blett, A. J. Gold, I. Goodman, L. H. Gunderson, B. M. Levinson, et al. 2006. "Ecological Thresholds: The Key to Successful Environmental Management or an Important Concept with no Practical Application?" *Ecosystems* 9: 1–13.
- Haltuch, M. A., K. F. Johnson, N. Tolimieri, M. S. Kapur, and C. Castillo-Jordan. 2019. *Status of the Sablefish Stock in US Waters in 2019* 398. Portland, OR: Pacific Fisheries Management Council.
- Hillebrand, H., I. Donohue, W. S. Harpole, D. Hodapp, M. Kucera, A. M. Lewandowska, J. Merder, J. M. Montoya, and J. A. Freund. 2020. "Thresholds for Ecological Responses to Global Change Do Not Emerge from Empirical Data." *Nature Ecology & Evolution* 4: 1502–9.
- Hooten, M. B., and N. T. Hobbs. 2015. "A Guide to Bayesian Model Selection for Ecologists." *Ecological Monographs* 85: 3–28.
- Hunsicker, M. E., C. V. Kappel, K. A. Selkoe, B. S. Halpern, C. Scarborough, L. Mease, and A. Amrhein. 2016. "Characterizing Driver–Response Relationships in Marine Pelagic Ecosystems for Improved Ocean Management." *Ecological Applications* 26: 651–663.
- Indivero, J., S. Anderson, L. Barnett, T. E. Essington, and E. Ward. 2024. "Estimating a Physiological Threshold to Oxygen and Temperature from Marine Monitoring Data Reveals Challenges and Opportunities for Forecasting Distribution Shifts." *Ecography*: in press. <https://doi.org/10.1111/ecog.07413>
- Jacobsen, N. S., K. N. Marshall, A. M. Berger, C. Grandin, and I. G. Taylor. 2022. "Climate-Mediated Stock Redistribution Causes Increased Risk and Challenges for Fisheries Management." *ICES Journal of Marine Science* 79: 1120–32.
- Kelly, R. P., A. L. Erickson, L. A. Mease, W. Battista, J. N. Kittinger, and R. Fujita. 2015. "Embracing Thresholds for Better Environmental Management." *Philosophical Transactions of the Royal Society, B: Biological Sciences* 370: 20130276.
- Krnjajić, M., and D. Draper. 2014. "Bayesian Model Comparison: Log Scores and DIC." *Statistics & Probability Letters* 88: 9–14.
- Kuha, J. 2004. "AIC and BIC: Comparisons of Assumptions and Performance." *Sociological Methods & Research* 33: 188–229.
- Large, S. I., G. Fay, K. D. Friedland, and J. S. Link. 2013. "Defining Trends and Thresholds in Responses of Ecological Indicators to Fishing and Environmental Pressures." *ICES Journal of Marine Science* 70: 755–767.
- Londschien, M., P. Bühlmann, and S. Kovács. 2023. "Random Forests for Change Point Detection." *Journal of Machine Learning Research* 24: 1–45.
- Malick, M. J., M. E. Hunsicker, M. A. Haltuch, S. L. Parker-Stetter, A. Berger, and K. Marshall. 2020. "Relationships between Temperature and Pacific Hake Distribution Vary across Latitude and Life-History Stage." *Marine Ecology Progress Series* 639: 185–197.
- Malick, M. J., M. E. Hunsicker, M. A. Haltuch, S. L. Parker-Stetter, K. N. Marshall, J. E. Pohl, A. M. Berger, S. A. Siedlecki, S. Gauthier, and A. J. Hermann. 2024. "Spatially Varying Effects of the California Undercurrent on Pacific Hake Distribution." *Canadian Journal of Fisheries and Aquatic Sciences* 81: 154–165.
- McClatchie, S. 2014. *Regional Fisheries Oceanography of the California Current System: The CalCOFI Program*. Dordrecht: Springer Netherlands.
- McHenry, J., H. Welch, S. E. Lester, and V. Saba. 2019. "Projecting Marine Species Range Shifts from Only Temperature Can Mask Climate Vulnerability." *Global Change Biology* 25: 4208–21.
- Morrison, W. E., S. A. Oakes, M. A. Karp, M. H. Appelman, and J. S. Link. 2024. "Ecosystem-Level Reference Points: Moving toward Ecosystem-Based Fisheries Management." *Marine and Coastal Fisheries* 16: e10285.
- Muhling, B. A., S. Brodie, J. A. Smith, D. Tommasi, C. F. Gaitan, E. L. Hazen, M. G. Jacox, T. D. Auth, and R. D. Brodeur. 2020. "Predictability of Species Distributions Deteriorates under Novel Environmental Conditions in the California Current System." *Frontiers in Marine Science* 7: 589.
- Munch, S. B., T. L. Rogers, and G. Sugihara. 2023. "Recent Developments in Empirical Dynamic Modelling." *Methods in Ecology and Evolution* 14: 732–745.
- Phillips, E. M., D. Chu, S. Gauthier, S. L. Parker-Stetter, A. O. Shelton, and R. E. Thomas. 2022. "Spatiotemporal Variability of Euphausiids in the California Current Ecosystem: Insights from a Recently Developed Time Series." *ICES Journal of Marine Science* 79: 1312–26.
- Phillips, E. M., M. J. Malick, S. Gauthier, M. A. Haltuch, M. E. Hunsicker, S. L. Parker-Stetter, and R. E. Thomas. 2023. "The Influence of Temperature on Pacific Hake co-Occurrence with Euphausiids in the California Current Ecosystem." *Fisheries Oceanography* 32: 267–279.
- Phillips, M. L., C. Lauria, T. Spector, J. B. Bradford, C. Gehring, B. B. Osborne, A. Howell, et al. 2024. "Trajectories and Tipping Points of piñon–Juniper Woodlands after Fire and Thinning." *Global Change Biology* 30: e17149.
- Poikane, S., M. G. Kelly, G. Várbiro, G. Borics, T. Erős, S. Hellsten, A. Kolada, et al. 2022. "Estimating Nutrient Thresholds for Eutrophication Management: Novel Insights from Understudied Lake Types." *Science of the Total Environment* 827: 154242.
- Puerta, P., A. Quetglas, and M. Hidalgo. 2016. "Seasonal Variability of Cephalopod Populations: A Spatio-Temporal Approach in the Western Mediterranean Sea." *Fisheries Oceanography* 25: 373–389.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna: R Foundation for Statistical Computing.
- Recknagel, F., I. Ostrovsky, H. Cao, T. Zohary, and X. Zhang. 2013. "Ecological Relationships, Thresholds and Time-Lags Determining Phytoplankton Community Dynamics of Lake Kinneret, Israel Elucidated by Evolutionary Computation and Wavelets." *Ecological Modelling* 255: 70–86.
- Samhouri, J. F., K. S. Andrews, G. Fay, C. J. Harvey, E. L. Hazen, S. M. Hennessey, K. Holsman, et al. 2017. "Defining Ecosystem Thresholds for Human Activities and Environmental Pressures in the California Current." *Ecosphere* 8: e01860.
- Samhouri, J. F., P. S. Levin, and C. H. Ainsworth. 2010. "Identifying Thresholds for Ecosystem-Based Management." *PLoS One* 5: e8907.
- Satterthwaite, W. H., K. S. Andrews, B. J. Burke, J. L. Gosselin, C. M. Greene, C. J. Harvey, S. H. Munsch, M. R. O'Farrell, J. F. Samhouri, and K. L. Sobocinski. 2020. "Ecological Thresholds in Forecast Performance for Key United States West Coast Chinook Salmon Stocks." *ICES Journal of Marine Science* 77: 1503–15.
- Schmidt, J. O., S. J. Bograd, H. Arrizabalaga, J. L. Azevedo, S. J. Barbeaux, J. A. Barth, T. Boyer, et al. 2019. "Future Ocean Observations to Connect Climate, Fisheries and Marine Ecosystems." *Frontiers in Marine Science* 6: 550.

- Schreiber, S. J. 2003. "Allee Effects, Extinctions, and Chaotic Transients in Simple Population Models." *Theoretical Population Biology* 64: 201–9.
- Selkoe, K. A., T. Blenckner, M. R. Caldwell, L. B. Crowder, A. L. Erickson, T. E. Essington, J. A. Estes, et al. 2015. "Principles for Managing Marine Ecosystems Prone to Tipping Points." *Ecosystem Health and Sustainability* 1: 1–18.
- Shotwell, S. K., K. Blackhart, C. Cunningham, E. Fedewa, D. Hanselman, K. Aydin, M. Doyle, et al. 2023. "Introducing the Ecosystem and Socioeconomic Profile, A Proving Ground for Next Generation Stock Assessments." *Coastal Management* 51: 319–352.
- Simpson, G. 2023. "gratia: Graceful ggplot-Based Graphics and Other Functions for GAMs Fitted using mgcv." R Package Version 0.8.1.46. <https://gavinsimpson.github.io/gratia/>.
- Sonderegger, D. L., H. Wang, W. H. Clements, and B. R. Noon. 2009. "Using SiZer to Detect Thresholds in Ecological Data." *Frontiers in Ecology and the Environment* 7: 190–95.
- Spake, R., M. P. Barajas-Barbosa, S. A. Blowes, D. E. Bowler, C. T. Callaghan, M. Garbowski, S. D. Jurburg, et al. 2022. "Detecting Thresholds of Ecological Change in the Anthropocene." *Annual Review of Environment and Resources* 47: 797–821.
- Sutherland, C., D. Hare, P. J. Johnson, D. W. Linden, R. A. Montgomery, and E. Droge. 2023. "Practical Advice on Variable Selection and Reporting Using Akaike Information Criterion." *Proceedings of the Royal Society B: Biological Sciences* 290: 20231261.
- USGCRP. 2023. "Ecosystems, Ecosystem Services, and Biodiversity." In *Fifth National Climate Assessment*, edited by A. R. Crimmins, C. W. Avery, D. R. Easterling, K. E. Kunkel, B. C. Stewart, and T. K. Maycock. Washington, DC: U.S. Global Change Research Program. <https://doi.org/10.7930/NCA5.2023>.
- Wang, Y. 2024. "The Effect of Climate Change on Forest Fire Danger and Severity in the Canadian Boreal Forests for the Period 1976–2100." *Journal of Geophysical Research: Atmospheres* 129: e2023JD039118.
- Weber, E. D., and S. McClatchie. 2012. "Effect of Environmental Conditions on the Distribution of Pacific Mackerel (*Scomber japonicus*) Larvae in the California Current System." *Fishery Bulletin* 110: 85–97.
- Wood, S. N. 2011. "Fast Stable Restricted Maximum Likelihood and Marginal Likelihood Estimation of Semiparametric Generalized Linear Models." *Journal of the Royal Statistical Society, Series B: Statistical Methodology* 73: 3–36.
- Wooster, W. S., and C. I. Zhang. 2004. "Regime Shifts in the North Pacific: Early Indications of the 1976–1977 Event." *Progress in Oceanography* 60: 183–200.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Detmer, A. Raine, Eric J. Ward, Mary E. Hunsicker, Kelly S. Andrews, Michele Conrad, Bridget E. Ferriss, Elliott L. Hazen, et al. 2025. "Evaluating the Robustness of Generalized Additive Models as a Tool for Threshold Detection in Variable Environments." *Ecosphere* 16(3): e70117. <https://doi.org/10.1002/ecs2.70117>